Q **July 25, 2026** Sean Li

Wen Motors

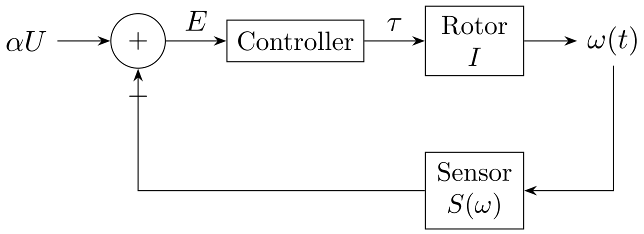
[10 pts]

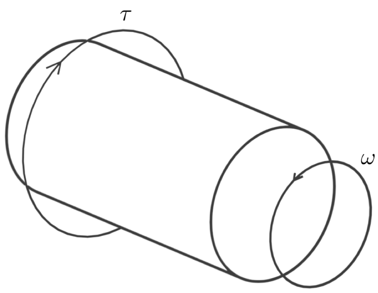
Dr. Wen owns a warehouse full of cheap DC motors. Although powerful, their angular velocities fluctuate unpredictably, making them unsuitable for precision applications. He wants to construct an ideal motor whose steady-state angular velocity depends only on the applied voltage:

$$\omega_{\mathrm{ss}} = \alpha U,$$

where U is the terminal voltage and $\alpha > 0$ is a constant.

To achieve this, Dr. Wen attaches a feedback controller to an ordinary motor. The resulting device, called a **Wen motor**, consists of a rotor, a speed sensor, a controller, and an actuator capable of applying corrective torque.





The controller continuously computes

$$E = \alpha U - S(\omega)$$

and applies the corrective torque

$$\tau = kE, \qquad k > 0.$$

Assume the rotor has moment of inertia I and no external torque acts on it.

Questions.

Q1. A naive controller. Dr. Wen first chooses the sensor

$$S(\omega) = \omega - \omega_0,$$

where $\omega_0 > 0$ is a constant. Using the control law above and Newton's Second Law in rotational form, show that for any constant voltage U, the rotor converges to a steady angular velocity, and determine the steady-state value.

Q2. The improved controller. Dr. Wen replaces the sensor with

$$S(\omega) = \ln\left(\frac{\omega}{\omega_0}\right).$$

He claims that logarithms are more suitable for controlling multiplicative quantities. Show that the motor converges again, and determine the steady-state relation between ω and U.